AURIS: An LLM-Driven Framework for Structured Mental Health Assessment Using Mixture-of-Experts Architecture

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Abstract. AURIS targets the mental health treatment gap with a retrieval-augmented, safety-aware conversational assessor. Built on the LLaMA 4 Scout mixture-of-experts model and a RAG pipeline, it adapts 14 validated questionnaires (178 items) across three phases: Rapport, Screening, and Snapshot. Using the ESHRO hybrid evaluation (automated metrics plus expert review), AURIS achieved a 0.76 composite score; human raters scored Safety at 1.00 and Empathy at 0.88, while automated relevance lagged at 0.53, highlighting that standard similarity metrics miss clinical alignment. The system is a preventive psychiatry prototype—an empathetic, low-friction first touchpoint rather than a therapy replacement. Future work includes domain-specific fine-tuning and clinically grounded automated metrics.